

1 Source CAD + request

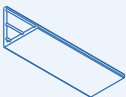
published instruction, held-out family

"Increase the box X dimension from 33.0 to 42.9."

program C (18 lines; no line hint)

```
0 import cadquery as cq
1 result = (
2   cq.Workplane("XY")
3   .box(33.0, 33.0, 83.0)
4   .faces(">Z").workplane()
5   .center(1.5, 1.5)
6   .rect(30.0, 30.0)
7   .cutThruAll()
8   .edges("|Z")
9   .

```



source solid, reference only:
the editor reads code + text only

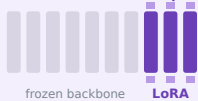
2 Patch editor (LoRA)

Qwen2.5-Coder-Instruct (fresh base)

predicted patch p (3B, this case) = gold

```
{ "edits": [ { "start": 3, "delete": 1, "insert": [ ".box(42.9, 33.0, 83.0)" ] } ] }
```

C, u



adapters on 7 attention/MLP projections
in the last 8 (1.5B) or 12 (3B) blocks;
loss on patch tokens only

size	precision, GPU	train	strict
1.5B	MLX 4-bit, M5	492	12/125
1.5B	BF16, H100	492	15/125
3B	NF4, T4	2,540	39/252

strict: held-out volume IoU ≥ 0.99999

↑ fine-tune

3 Validate + execute

- ✓ schema: start, delete, insert
- ✓ ordered, disjoint, in range
- ✓ original-line indices
- ✓ patched program parses

executed program $\hat{C} = \mathcal{E}(C, p)$

3 - .box(33.0, 33.0, 83.0)

3 + .box(42.9, 33.0, 83.0)

restricted CadQuery run

allow-listed imports, no file I/O,
45 s per case

solid V = 41,995.2 mm³

✗ otherwise: unverified

no retry, no repair. Held-out 3B:
74 invalid patches, 32 execution
errors, 1 timeout

↓ prediction + target solids, if a contract exists

4 Linked SVG views

views of the predicted solid

front



top



right



isometric



one solid drives all four views;
each view is autoscaled

5 Independent scoring

vs. a hash-pinned reference solid

$$\text{IoU} = \frac{V(\hat{B} \cap B^*)}{V(\hat{B}) + V(B^*) - V(\hat{B} \cap B^*)}$$

this case: IoU = 1.000000001

$\hat{V} = V^* = 41,995.194 \text{ mm}^3$

program AST = released target ✓

STRICT MATCH

final 3B adapter, held-out set

held-out tasks 252

applicable patch 173

executed + scored 139

IoU ≥ 0.95 52

IoU ≥ 0.99999 39

6 reference errors and 1 timeout
stay in n = 252

all 252 tasks ↓

D Training data + leakage-safe split

748

BenchCAD
edit pairs

710

mechanical
(-38 pipe/circuit)

533/50/127

family-component
split

492/44/126

fit 3,072-token
context

100 part families, each kept whole (width ∝ pairs):



■ train 74 families ■ val 8 ■ test 18, incl. mounting angle

families sharing a numeric-masked AST template are merged before hash-ranked
assignment; each target is a comment-stripped line diff replayed to its target

3B run adds CAD-Editor sketch-extrude pairs: 2,540 training pairs in total

6 Conditional FEM audit

explicit contract only

test contract

units: mm, N, MPa
E 210 GPa, ν 0.3
fixed: bottom 5 %
1 N load, top 5 %
hashed before solve

*hypothetical
material; no
safety rating*

solve $Kq = f$



33,871 nodes
von Mises, p95-clipped

numerical checks

force balance 2e-14 ✓
energy identity 1e-14 ✓
mesh volume 0.16 % ✓
refinement ΔC 0.06 % ✓
C vs. target 0.0011 % ✓
C: compliance; tol. 3 / 5 / 10 %

*geometry is checked first:
wrong shapes can match C*

patch checks · execution · IoU · AST · FEM
evaluation only: never a reward, loss, retry or repair

7 Recorded outcome

✓ VERIFIED (diagnostic)

strict geometry + FEM checks:
4 of 6 illustrated 3B cases

✗ UNVERIFIED, kept

patch, execution or mesh
failure; no gold edit,
no retry, no repair

What trains
vs. what checks

$\mathcal{L}(\theta) = -\sum_{i \in p} \log p_{\theta}(y_i | y_{< i}, C, u)$
the only training signal: next-token loss on patch tokens

Every held-out task is
reported, failures included.